PADHAKU
Exploration · Class 9 Science



CHAPTER 7

Work, Energy, and Simple Machines

NCERT Exercise Solutions

- Think It Over — 3 Q
- Pause and Ponder · 7.1 Work Done by a Constant Force — 2 Q
- Pause and Ponder · 7.2 The Work-Energy Theorem — 1 Q
- Pause and Ponder · 7.4.1 Kinetic energy — 2 Q
- Activity 7.1: Let us investigate — 1 Q
- Pause and Ponder · 7.4.2 Potential energy — 1 Q
- Activity 7.2: Let us experiment — 1 Q
- Pause and Ponder · 7.4.3 Conservation of mechanical energy — 2 Q
- Activity 7.3: Let us experiment — 1 Q
- Pause and Ponder · 7.6.2 Inclined plane — 2 Q
- Activity 7.4: Let us investigate — 1 Q

✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

Independent study aid by Beginers Point. Not an official NCERT publication.

🤔 Think It Over

1 **What will be the magnitude of velocity of the child at the bottom of the blue slide?**

✔ ANSWER

All the potential energy at the top changes into kinetic energy at the bottom (ignoring friction). ½ mv^2 = mgh, so **v = √(2gh)**, where h is the height of the top of the blue slide.

2 **Will two children of different masses reach the bottom of the same slide with the same velocity?**

✔ ANSWER

Yes (ignoring friction). v = √(2gh) does not contain the mass, so it depends only on the height of the slide.

3 **Which of the slides will result in the largest magnitude of velocity for the child at its bottom?**

💡 SAMPLE ANSWER

The speed at the bottom depends only on the height from which the child starts, not on the shape of the slide.

So the slide that starts from the **greatest height** gives the largest speed. Slides starting from the same height give the same speed (ignoring friction). (check the picture in your book)

⭐ Pause and Ponder · 7.1 Work Done by a Constant Force

1 **A weightlifter is holding a barbell steady in her hands (Fig. 6.8). Is she doing any work on the barbell while holding it steady?**

✔ ANSWER

No. The barbell does not move, so displacement s = 0.

Work = F × s = F × 0 = **0**. (She still feels tired because her muscles use up body energy.)

2 **Is the work done by friction on the stack of coins that travels on a rough surface (Fig. 6.13c) — positive, negative or zero?**

✔ ANSWER

Negative. Friction acts opposite to the direction of motion (displacement) of the coins, so it does negative work and slows them down.

Pause and Ponder · 7.2 The Work-Energy Theorem

3 **When you pedal a bicycle on a flat road, your muscles supply energy. In what forms does this muscular energy appear as you ride?**

✔ ANSWER

- **Kinetic energy** of the moving bicycle and rider (the main part).
- **Thermal energy (heat)** due to friction in the chain, wheels, road and air resistance; your body also gets warm.
- A little **sound energy** (noise of the chain and tyres).

Pause and Ponder · 7.4.1 Kinetic energy

4 **Two objects A and B of mass m and 4m have the same kinetic energy. What is the ratio of the magnitude of velocities of A and B?**

✔ ANSWER

$\frac{1}{2}\, m\, v_A^2 = \frac{1}{2}\, (4m)\, v_B^2$

$v_A^2 = 4\, v_B^2 \rightarrow v_A = 2\, v_B$

$v_A : v_B = 2 : 1$

5 **Does the kinetic energy of an object which moves with constant velocity change with its position?**

✔ ANSWER

No. $K = \frac{1}{2} mv^2$ depends only on mass and velocity. If the velocity is constant, the kinetic energy stays the same at every position.

Activity 7.1: Let us investigate

Activity 7.1 **Drop a heavy ball into loose sand from about 1 m, then from 2 m (twice, at different spots). Is a depression created? Why? Compare the depths of the depressions.**

✔ ANSWER

Observation: Yes, the ball makes a depression (dent) in the sand. The depression is **deepest** when the ball falls from the greatest height and **shallowest** from the lowest height.

Why: The raised ball has energy; on hitting the sand it does work to push the sand aside.

Conclusion: More work is needed to raise the ball higher, so the higher the ball, the greater its potential energy.

⭐ Pause and Ponder · 7.4.2 Potential energy

6 **Does the potential energy of an object near the surface of the Earth change if it moves with constant velocity in the horizontal direction? What if the object is gradually raised in the vertical direction?**

✔ ANSWER

- Moving horizontally: the height h does not change, so U = mgh **does not change**.
- Raised vertically: h increases, so the potential energy **increases** (by mg × gain in height).

🧪 Activity 7.2: Let us experiment

Activity 7.2 **Release a pendulum bob from point P at the level of a horizontal line drawn on paper. Does the bob almost reach the level of the horizontal line at the extreme points of the first few oscillations?**

✔ ANSWER

Observation: Yes, on the other side (point R) the bob rises almost to the same horizontal line.

- At P and R: only potential energy (bob momentarily at rest).

- At Q (lowest point): only kinetic energy (fastest).

Conclusion: Mechanical energy of the bob stays (almost) constant. It slowly stops because of friction at the support and air resistance.

⭐ Pause and Ponder · 7.4.3 Conservation of mechanical energy

7 **For the situation depicted in Fig. 7.19 (object of mass m dropped from height h), calculate the mechanical energy of the ball just before it hits the ground and show that even at this position, it is mgh.**

✔ ANSWER

Just before the ground, height = 0, so potential energy = mg × 0 = 0.

Using $v^2 = u^2 + 2gh$ with u = 0: $v^2 = 2gh$.

Kinetic energy = ½ mv^2 = ½ m × 2gh = mgh.

Mechanical energy = 0 + mgh = **mgh** (same as at the start).

8 **In the ball roller coaster (Fig. 7.22), a ball is released from the highest point. Describe how the kinetic energy and potential energy change at points A, B and C. Why do subsequent points, such as C, D and E, usually have lower heights? Could it have**

anything to do with the energy lost due to friction?

✔ ANSWER

- From the start to **A**: the ball loses some height, so part of its potential energy becomes kinetic energy.
- **B** (lowest point): potential energy is least, kinetic energy is greatest (fastest).
- **C** (rises again): kinetic energy changes back into potential energy, so the ball slows down.

Yes. Friction and air resistance convert part of the mechanical energy into heat and sound at every step. With less mechanical energy left, the ball can rise only to smaller heights, so C, D, E must be lower than the earlier points. (check the picture in your book)

🧪 Activity 7.3: Let us experiment

Activity 7.3 **Lift a cart (with a spring balance) vertically to a height of about 0.5 m, then pull it up a plank to the same height. Is the force smaller along the plank? How does the force change when the plank is made less steep?**

✔ ANSWER

Observation: The spring balance reading along the plank is **smaller** than when lifting vertically. When the plank is less steep (longer), the reading becomes **even smaller**.

Conclusion: An inclined plane reduces the force (effort), but the force must be applied over a longer distance. The work done (force × distance) stays the same.

⭐ Pause and Ponder · 7.6.2 Inclined plane

9 **Explain why roads on hills are built to wind around in gentle slopes rather than going straight up.**

✔ ANSWER

A winding road is a long inclined plane with a gentle slope. Mechanical advantage = L/h, so a longer road for the same height needs a **smaller force** from the vehicle's engine to climb.

Vehicles can climb easily and safely, without slipping back. The work done is the same, but it is spread over a longer distance.

10 **To reach a higher floor, we find climbing an inclined ladder easier in comparison to climbing a vertical ladder (Fig. 7.30). Explain why.**

✔ ANSWER

An inclined ladder acts like an inclined plane. We cover a longer distance to reach the same height, so the force needed at each step is **less** than lifting our whole weight straight up on a vertical ladder.

The work done (mgh) is the same in both cases.

🧪 Activity 7.4: Let us investigate

Activity 7.4 **Place a 30 cm scale over a pencil near one end, put a stapler on the short side and erasers on the long side. Does the stapler lift up?**

✔ ANSWER

Observation: With one or two erasers on the far end, the heavier stapler is lifted.

Conclusion: The scale works as a **lever** (pencil = fulcrum). A small effort on a long effort arm can lift a heavy load on a short load arm.

Activity 7.5: Let us experiment

Activity 7.5 **Make a beam balance with a scale and two paper cups. Keep 1 coin in the left pan at a fixed distance L_1. Put 1, 2, 4, 8 coins in the right pan and move it towards the centre until the beam balances. Complete Table 7.1.**

💡 SAMPLE ANSWER

Sample readings (left pan kept at 20 cm):

n_1 (effort)	L_1 (cm)	n_2 (load)	L_2 (cm)
1	20	1	20
1	20	2	10
1	20	4	5
1	20	8	2.5

Conclusion: The beam balances when $n_1 \times L_1 = n_2 \times L_2$, i.e. **effort × effort arm = load × load arm**. Doubling the load halves its distance from the fulcrum.

Pause and Ponder · 7.6.3 Lever

11 **Why is it easier to open the lid of a can by using a spoon as shown in Fig. 7.35?**

✔ ANSWER

The spoon works as a **lever**. The rim of the can is the fulcrum, the lid is the load (very close to the fulcrum) and we press the far end of the handle (long effort arm).

Mechanical advantage = effort arm ÷ load arm is large, so a small effort produces a large force on the lid.

12 **Why do you push an object closer to the scissors' fulcrum when you want to cut an object which is hard?**

✔ ANSWER

Moving the object closer to the fulcrum makes the **load arm shorter**. Then effort arm ÷ load arm becomes larger.

So the same effort by the hand produces a **larger force** on the hard object, and it gets cut easily.

13 **Many designs of perpetual machines have been proposed but none actually work. Why do all real machines eventually slow down and stop? Explain in terms of work and energy.**

✔ ANSWER

In every real machine, **friction** and air resistance do negative work. They change part of the mechanical energy into heat (and sound), which cannot be fully used again.

A machine cannot create energy. If no fresh energy (fuel, electricity, push) is supplied, its mechanical energy keeps decreasing, so it slows down and stops.

What If ... **What if it were possible to build a perpetual motion machine, which once started, could continue doing useful work forever, without any fuel or electricity?**

💡 **SAMPLE ANSWER**

It would produce energy from nothing. This would break the law of conservation of energy, so such a machine is **not possible**.

If it existed, we would never need fuel or electricity — but in reality friction always takes away some energy.

⭐ Revise, Reflect, Refine

1 **State whether True or False.**

(i) Work is said to be done when a force is applied, even if the object does not move.

(ii) Lifting a bucket vertically upward results in positive work done on the bucket.

(iii) The SI unit for both work and energy is joule (J).

(iv) A motionless stretched rubber band has kinetic energy.

(v) Energy can change from one form to another.

✔ ANSWER

(i) **False** — if there is no displacement, no work is done.

(ii) **True** — force and displacement are both upward.

(iii) **True**

(iv) **False** — it is not moving; it has (elastic) potential energy.

(v) **True**

2 **Fill in the blanks.**

(i) Work done = _____ × _____ (in the direction of force).

(ii) 1 joule of work is done when a force of _____ newton displaces an object by 1 metre in the direction of the force.

(iii) The expression for kinetic energy of a body of mass m and velocity v is _____.

(iv) The potential energy of an object of mass m at a small height h from the Earth's surface is _____.

(v) Power is defined as the _____ at which work is done.

✔ ANSWER

(i) **force × displacement**

(ii) **1**

(iii) **½ mv²**

(iv) **mgh**

(v) **rate**

3 **When a ball thrown upwards reaches its highest point, tick which of the following statement(s) are correct?**

(i) The force acting on the ball is zero.

(ii) The acceleration of the ball is zero.

(iii) Its kinetic energy is zero.

(iv) Its potential energy is maximum.

✔ ANSWER

Correct: **(iii) and (iv)**.

- (i) Wrong: gravity (mg) still acts on the ball.
- (ii) Wrong: acceleration is g (downward), not zero.
- (iii) Right: the ball is momentarily at rest, so v = 0.
- (iv) Right: the ball is at its greatest height.

4 **Identify the energy transformation in: (i) a truck moving uphill, (ii) unwinding of a watch spring, (iii) photosynthesis in green leaves, (iv) water flowing from a dam, (v) burning of a matchstick, (vi) explosion of a fire cracker, (vii) speaking into a microphone, (viii) a glowing electric bulb, and (ix) a solar panel.**

✔ ANSWER

(i) Chemical energy (fuel) → kinetic + potential energy (and some heat)

(ii) Elastic potential energy → kinetic energy

(iii) Light energy → chemical energy (food)

(iv) Potential energy → kinetic energy

(v) Chemical energy → heat (thermal) + light energy

(vi) Chemical energy → heat, light, sound and kinetic energy

(vii) Sound energy → electrical energy

(viii) Electrical energy → light + heat energy

(ix) Light (solar) energy → electrical energy

5 **A student is slowly lifted straight up in an elevator from the ground to the top floor of a building. Later, the same student climbs the staircase to the top. h = 72.5 m, g = 10 m s^{-2}, m = 50 kg.**

(i) Find the gain in the potential energy if the student is lifted straight up to the top.

(ii) Find the gain in the potential energy when the student climbs the stairs to the same top.

(iii) What do you conclude about the dependence of the potential energy on the path taken?

✔ ANSWER

(i) Given m = 50 kg, g = 10 m s^{-2}, h = 72.5 m

U = mgh = 50 × 10 × 72.5 = **36250 J**

(ii) The height gained is the same, 72.5 m, so gain in potential energy = **36250 J**.

(iii) Potential energy depends only on the **height** (initial and final positions), **not on the path** taken.

6 **A crane lifts a mass m to the 10th floor of a building in a certain time. It then raises the same mass to the 20th floor of the same building in double the time. How much more energy and power are required? Assume that the height of all floors is equal.**

✔ ANSWER

Let height of 10 floors = h and time = t (both lifts start from the ground).

Energy: $E_1 = mgh$; $E_2 = mg(2h) = 2E_1$. So the energy needed is **double** (E_1 more).

Power: $P_1 = E_1/t$; $P_2 = 2E_1/2t = E_1/t = P_1$.

So **no more power** is needed; the power stays the same.

7 **Which factors determine the energy required to raise a flag from the ground to the top of a tall flagpole using a pulley? Does raising the flag slowly or quickly change the amount of work done? If the speed at which the flag is raised is doubled, how does the power requirement change?**

✔ ANSWER

- Energy needed = mgh, so it depends on the **mass (weight) of the flag** and the **height of the pole** (and a little on friction in the pulley).
- **No.** The work done (mgh) is the same whether the flag is raised slowly or quickly.
- If the speed is doubled, the time becomes half. P = W/t, so the **power needed doubles**.

8 **A man of mass 60 kg rides a scooter of mass 100 kg and accelerates it to a velocity v. The next day, his son of mass 40 kg joins him. If the scooter reaches the same speed on both days in the same time interval, what is the ratio of the fuel used on the two days? (No losses.)**

✔ ANSWER

Day 1 mass = 60 + 100 = 160 kg; Day 2 mass = 160 + 40 = 200 kg.

Fuel used ∝ kinetic energy gained = ½ Mv^2 (same v).

Ratio = ½ × 160 × v^2 : ½ × 200 × v^2 = 160 : 200

Fuel ratio (Day 1 : Day 2) = 4 : 5

9 **On a seesaw with sliding seats, a child sits on one side and an adult on the other. The adult weighs twice the child, yet the seesaw is balanced. Draw a figure showing the distances from the fulcrum.**

💡 SAMPLE ANSWER

Using load × load arm = effort × effort arm:

$W \times d_1 = 2W \times d_2 \rightarrow d_1 = 2\,d_2$

So the child sits **twice as far** from the fulcrum as the adult.

Figure (example): Child (W) ——— 2 m ——— ▲ fulcrum —— 1 m —— Adult (2W)

10 **A ball of mass 2 kg is thrown up with a velocity of 20 m s^{-1}.**

(i) Identify the sign of the work done by gravity on the ball during its upward motion and its downward motion.

(ii) If the ball reaches a height of 19.4 m, how much work was done by air resistance (g = 10 m s^{-2})?

✔ ANSWER

(i) Going up: gravity is downward, displacement upward → work by gravity is **negative**.

Coming down: both downward → work by gravity is **positive**.

(ii) Initial KE = ½ × 2 × 20^2 = 400 J (initial PE = 0)

At the top: KE = 0, PE = mgh = 2 × 10 × 19.4 = 388 J

Work by air resistance = change in mechanical energy = 388 J − 400 J = **−12 J**

11 **A 10.0 kg block moves on a horizontal floor with negligible friction. A variable force (Fig. 7.37) acts in its direction of motion from 0 m to 4 m. If the block had a kinetic energy of 180 J at 0 m, find its speed (i) at 0 m, and (ii) at 4 m. Does the block have negative acceleration in any portion of its motion?**

Graph (Fig. 7.37): force rises from 0 to 50 N between 0 and 1 m, stays 50 N from 1 m to 3 m, falls to 0 N at 4 m.

✔ ANSWER

(i) $\frac{1}{2} mv^2 = 180$ J → $v^2 = 2 \times 180 \div 10 = 36$ → $v =$ **$6\ m\ s^{-1}$**

(ii) Work done = area under the graph:

$= \frac{1}{2} \times 1 \times 50 + 2 \times 50 + \frac{1}{2} \times 1 \times 50 = 25 + 100 + 25 = 150$ J

KE at 4 m = 180 + 150 = 330 J

$v^2 = 2 \times 330 \div 10 = 66$ → $v = \sqrt{66} \approx$ **$8.1\ m\ s^{-1}$**

No negative acceleration: the force is always in the direction of motion (never negative), so the block keeps speeding up (acceleration is zero only at 0 m and 4 m where F = 0).

12 **The gravitational attraction on the surface of the Moon is about 1/6th of that on the Earth. An astronaut can throw a ball up to a height of 8 m on the Earth. How far up will the ball thrown with the same upward velocity travel on the Moon?**

✔ ANSWER

At the top all KE becomes PE: ½ mv^2 = mgh → h = $v^2/(2g)$.

Same v, so h is inversely proportional to g.

g on Moon = g/6, so h on Moon = 6 × 8 m = **48 m**

13 **A 1000 kg car moves at a constant speed. The driver sees an obstruction and brakes to a stop. Fig. 7.38 (speed–time): speed 35 m s^{-1} from A (0 s) to B (1 s), then falls uniformly to 0 at C (3 s).**

(i) Describe how the car moves between positions A and B.

(ii) Calculate the kinetic energy of the car at A.

(iii) State the work done by the brakes in bringing the car to a halt between B and C.

(iv) What does the kinetic energy of the car transform into?

✔ ANSWER

(i) Between A and B the car moves with a **constant speed of 35 m s^{-1}** (zero acceleration) for 1 s — the time before the driver applies the brakes.

(ii) $K = \frac{1}{2} mv^2 = \frac{1}{2} \times 1000 \times 35^2 = 500 \times 1225 =$ **612500 J**

(iii) Work by brakes = change in KE = 0 − 612500 J = **−612500 J** (negative work).

(iv) Mainly into **heat (thermal energy)** in the brakes and tyres, and some **sound**.

14 **The potential energy–displacement graph of a 0.5 kg ball on a frictionless track is shown in Fig. 7.39. At O the velocity is 0 m s^{-1} and potential energy is 30 J. Calculate the velocity of the ball at P, Q and R.**

From the graph: PE at P = 20 J, at Q = 30 J, at R = 40 J.

💡 **SAMPLE ANSWER**

Total mechanical energy = 30 J + 0 = 30 J (constant, no friction).

At P: KE = 30 − 20 = 10 J → $v^2 = 2 \times 10 \div 0.5 = 40$ → $v = \sqrt{40} \approx$ **6.3 m s^{-1}**

At Q: KE = 30 − 30 = 0 → v = **0 m s^{-1}** (the ball stops here and turns back).

At R: PE needed = 40 J, more than the total 30 J. So the ball **cannot reach R**; it never gets beyond Q.

15 **A coconut of mass 1.5 kg falls from the top of a 10 m tall coconut tree onto wet sand and comes to rest, making a depression.**

(i) Calculate the velocity of the coconut just before it hits the sand.

(ii) If the average resistive force of sand is 3000 N and all the coconut's energy is used to make the depression, find its depth. (g = 10 m s^{-2})

✔ ANSWER

(i) ½ mv^2 = mgh → v = √(2gh) = √(2 × 10 × 10) = √200 ≈ **14.1 m s^{-1}**

(ii) Energy of coconut = mgh = 1.5 × 10 × 10 = 150 J

Work done by sand = force × depth → 3000 N × d = 150 J

d = 150 ÷ 3000 = 0.05 m = **5 cm**

⭐ The Journey Beyond

1 **Make a pen-refill shooter with a rubber band. Stretch and release the rubber band by different amounts. Is there a relationship between the stretch and the distance travelled?**

💡 SAMPLE ANSWER

Yes. The **more** the rubber band is stretched, the more elastic potential energy it stores, so the refill gets more kinetic energy and travels **farther**.

2 **Construct one or more simple machines (lever, pulley, inclined plane) from easily available materials. Use your model to lift a small load, measure the effort and load, and calculate the mechanical advantage.**

💡 **SAMPLE ANSWER**

Example: a ruler lever on a pencil fulcrum lifts a load of 2 N with an effort of 0.5 N.

Mechanical advantage = load ÷ effort = 2 ÷ 0.5 = **4** (here the effort arm is about 4 times the load arm).

3 **Use PhET simulations (Energy Skate Park, Energy Forms and Changes, Pendulum Lab, Masses and Springs) to explore how different forms of energy change as mass, height and friction are varied.**

💡 **SAMPLE ANSWER**

- Greater height → more potential energy at the top and more kinetic energy at the bottom.
- Mass changes the energies but not the speed at the bottom (without friction).

- With friction, total mechanical energy decreases and thermal energy increases.

🎉 **Chapter 7 done! Revise the answers once more tomorrow.**